

EXERCISES

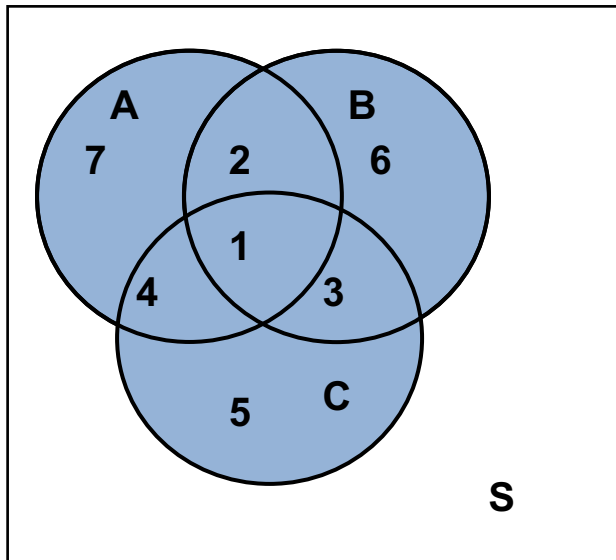
1. The file SavingsRate contains the yields for one-year Deposits and five-year Deposits, for 23 banks in the United States, as of April 15, 2015.

For each type of account:

- a. Compute the first, second, and third quartiles, and the interquartile range.
- b. List the five-number summary.
- c. Construct a boxplot (use Excel) and describe its shape.

EXERCISES

2. Calculate:



a. $A \cap B$

b. $B \cap C$

c. $A \cup C$

d. $A \cap B'$

e. $A \cap B \cap C$

f. $(A \cup B) \cap C'$

EXERCISES

3. Do males or females feel more tense or stressed out at work? A survey of employed adults conducted online by Harris Interactive on behalf of the American Psychological Association revealed the following:

FELT TENSE OR STRESSED OUT AT WORK

GENDER	Yes	No
Male	244	495
Female	282	480

Source: Data extracted from “The 2013 Work and Well-Being Survey,” American Psychological Association and Harris Interactive, March 2013, p. 5, bit.ly/1JGcPf.

- Give an example of a simple event.
- Give an example of a joint event.
- What is the complement of “Felt tense or stressed out at work”?
- Are being Male and feeling tense or stressed out at work mutually exclusive events?

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- e. Given that the employed adult felt tense or stressed out at work, what is the probability that the employed adult was a male?
- f. Given that the employed adult is male, what is the probability that he felt tense or stressed out at work?
- g. Explain the difference in the results in (a) and (b).
- h. Is feeling tense or stressed out at work and gender independent?

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4. A car repair is either on time or late and either satisfactory or unsatisfactory. If a repair is made on time, then there is a probability of 0.85 that is satisfactory. There is a probability of 0.77 that a repair will be made on time. What is the probability that a repair is made on time and is satisfactory?

EXERCISES

5. A college found that 30% of its students left after the 1st year without completing the introductory statistics course. Assume that 20 students registered for the course and the probability of leaving the college (withdrawals) follows a binomial distribution, compute:

- a. The expected number of withdrawals
- b. The probability that exactly 3 students withdraw
- c. The probability that more than 3 students withdraw

EXERCISES

6. In a local airline reservation desk, phone calls come in at the rate of 48 calls every hour. Assume the distribution of the calls follows a Poisson Distribution. Calculate the probability of
- a) receiving 3 calls in a 5-minute interval of time
 - b) receiving less than 3 calls in a 5-minute interval of time
 - c) receiving more than 3 calls in a 5-minute interval of time
 - b) receiving exactly 10 calls in 15 minutes

**Please submit your work
by sending it to
dwiardini@binus.edu
by 23:59 WIB on Fri, 25 Mar 2023**

**Name your work-file as
“ GSLC 1_LB09_your name.docx/xlsx ”**